

A Multi-Dataset Study of EEG Seizure Detection

Furqan Khalil

Institute of Management Sciences

MS Data Science

Instructor: Dr. Adnan Amin

Major Assignment

Abstract—This paper presents a systematic evaluation of logistic regression for binary seizure detection across three publicly available EEG datasets: BEED (8,000 samples, 16 features, 75% seizure), UCI Epileptic Seizure Recognition (11,500 samples, 178 features, 20% seizure), and Bonn EEG (4,097 samples, 4,097 features, 50% seizure). Two preprocessing pipelines are evaluated on each dataset: Pipeline A (RobustScaler / IQR clipping, ANOVA SelectKBest) and Pipeline B (statistical feature extraction from raw time-series, Z-score standardization, PCA). The study investigates L1, L2, and Elastic Net regularization, and four class-imbalance strategies (Baseline, SMOTE, Undersampling, Class Weight). Results show that preprocessing design is the dominant performance factor, Pipeline B's statistical feature extraction consistently outperforms raw dimensionality reduction, L2 regularization generalises most reliably, and imbalance handling interacts strongly with both dataset characteristics and regularization type.

Index Terms—logistic regression, regularization, EEG, seizure detection, class imbalance, SMOTE, preprocessing, PCA, ANOVA

I. INTRODUCTION

Automated seizure detection from electroencephalogram (EEG) signals is a clinically significant problem. False negatives — missed seizures — carry severe patient risk, while false positives increase clinical workload. The diversity of EEG recording setups produces datasets with fundamentally different characteristics: varying dimensionality, class balance, and signal structure.

This assignment applies logistic regression to three EEG datasets under a controlled experimental framework. The same seven-step protocol is applied to each dataset: (1) data loading and exploratory analysis, (2) preprocessing pipelines, (3) baseline modelling, (4) overfitting and underfitting analysis, (5) regularization study, (6) class imbalance handling, and (7) cross-dataset comparative analysis.

II. DATASETS

A. BEED Dataset

The Brain Electrical Activity Data (BEED) contains 8,000 samples with 16 pre-computed frequency-band features. The five original classes are binarised: classes 1–3 map to Seizure (1) and classes 4–5 to Non-Seizure (0), yielding a 75%/25% class split. No missing values are present.

B. UCI Epileptic Seizure Recognition

This dataset contains 11,500 samples, each representing 178 raw EEG time-step measurements recorded at 178.57 Hz. The

original five classes are binarised so that class 1 (eyes-open seizure activity) maps to Seizure and classes 2–5 to Non-Seizure, producing a severe 20%/80% imbalance. An identifier column (Unnamed) is dropped before any processing.

C. Bonn EEG Dataset

The Bonn dataset contains 4,097 samples with 4,097 raw time-point features per sample, representing complete single-channel EEG segments. After binarisation the dataset is balanced at approximately 50%/50%, but the extremely high dimensionality (4,097 features) makes direct feature-level modelling intractable.

Table I summarises the three datasets.

TABLE I: Dataset Summary

Dataset	Samples	Features	Seizure%	Imbalance
BEED	8,000	16	75%	Mild
UCI	11,500	178	20%	Severe
Bonn EEG	4,097	4,097	50%	Balanced

III. PREPROCESSING PIPELINES

Critical design principle: the train/test split is performed *before* any preprocessing. All scalers, selectors, and resamplers are fitted exclusively on training data and applied to the test set — preventing data leakage throughout the study.

A. Pipeline A — Feature Engineering Path

- 1) **RobustScaler / IQR Clipping:** RobustScaler (BEED, UCI) or IQR-based clipping suppresses EEG voltage spikes without discarding samples. Fitted on training data only.
- 2) **ANOVA SelectKBest:** Supervised feature selection retaining the top- k features by F-statistic computed on training labels ($k = 10$ for BEED, $k = 50$ for UCI, $k = 30$ for Bonn after prior extraction).

B. Pipeline B — Statistical Extraction + PCA

- 1) **Statistical Feature Extraction:** For UCI and Bonn (raw time-series), six statistical features are computed per sample: mean, std, peak-to-peak amplitude, IQR, mean absolute difference (activity), and mean energy. This reduces 178 or 4,097 dimensions to 6 interpretable features before PCA.
- 2) **Z-score Standardisation:** Centres each feature to zero mean and unit variance using training statistics.

- 3) **PCA (95% variance):** Retains the minimum number of components explaining 95% of variance. Fitted on training data only.

The key distinction is that Pipeline B applies *domain-aware* feature extraction before dimensionality reduction, rather than feeding raw time-series directly into PCA.

IV. BASELINE LOGISTIC REGRESSION

A logistic regression with default L2 regularisation ($C = 1.0$) is trained independently on each pipeline's output. For UCI and Bonn, `class_weight='balanced'` is used in the baseline because without it the 20% and 50% minority classes are ignored entirely by the optimiser.

TABLE II: Baseline Logistic Regression Results

Dataset	Pipeline	Acc.	F1	PR-AUC
BEED	Pipeline A	0.669	0.761	0.746
	Pipeline B	0.832	0.892	0.726
UCI	Pipeline A	0.537	0.284	0.371
	Pipeline B	0.952	0.886	0.955
Bonn	Pipeline A	0.750	0.444	0.541
	Pipeline B	0.887	0.727	0.682

Figures 1–3 show the baseline comparison bar charts for all three datasets. Pipeline B consistently outperforms Pipeline A, confirming that statistical feature extraction from raw time-series provides far more discriminative representations than ANOVA feature selection on scaled raw values.

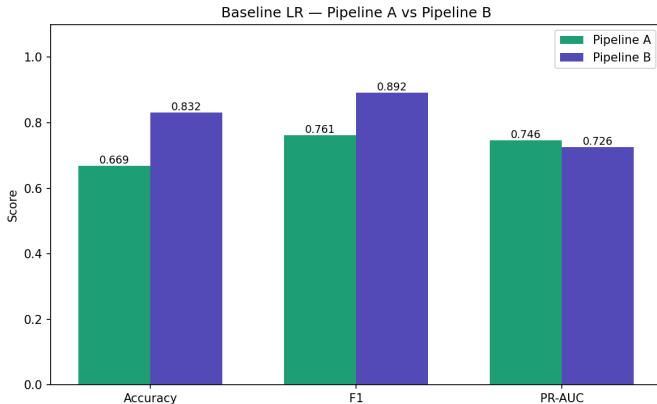


Fig. 1: Baseline LR — BEED: Pipeline A vs Pipeline B.

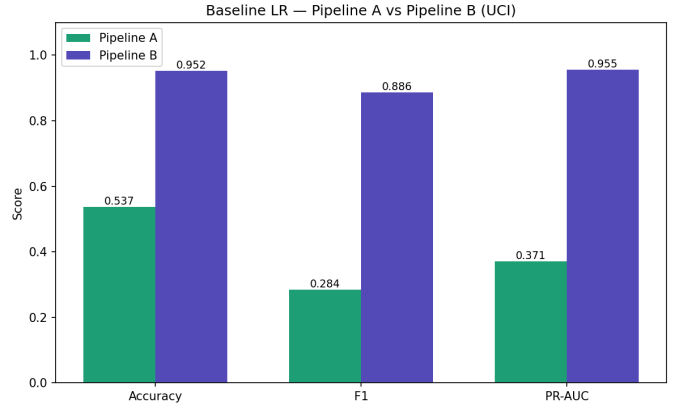


Fig. 2: Baseline LR — UCI Epileptic: Pipeline A vs Pipeline B.

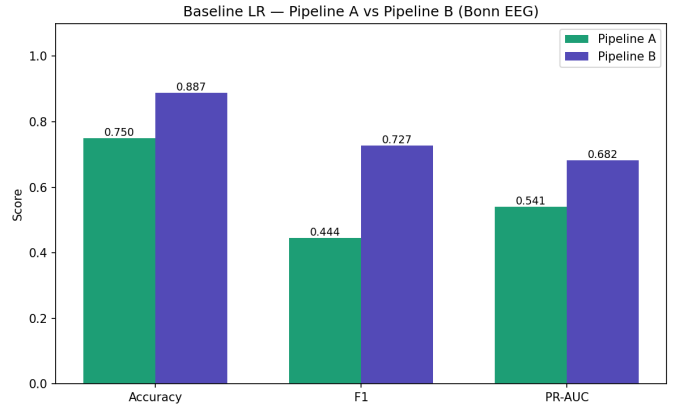


Fig. 3: Baseline LR — Bonn EEG: Pipeline A vs Pipeline B.

V. OVERFITTING AND UNDERFITTING ANALYSIS

Three scenarios are constructed on Pipeline A for each dataset:

Underfitting: Restricting the model to 5 features with $C = 0.0001$ produces high-bias behaviour. Both train and validation F1 are depressed and flat, with near-zero gap, confirming insufficient model capacity.

Baseline: The standard configuration (k selected features, $C = 1.0$) shows a healthy convergence pattern on learning curves — train F1 starts higher and validation F1 gradually closes the gap as sample size increases.

Overfitting: Setting $C = 1000$ (minimal regularisation) produces only a marginally wider train-validation gap. This is an expected and documented property of logistic regression: its global linear boundary cannot memorise individual training points, making it structurally resistant to overfitting compared to non-linear models such as decision trees or k -NN.

Pipeline B learning curves show a notably different pattern: all three scenarios (underfit, baseline, overfit) converge to high F1 (≈ 0.84 – 0.90) with near-zero train-validation gap. This confirms that the six statistical features extracted in Pipeline B are so discriminative that regularisation strength has minimal effect on generalisation.

Validation curves (sweeping C over $[10^{-4}, 10^4]$) confirm that $C = 1.0$ lies in the optimal zone for all three datasets on Pipeline A.

VI. REGULARIZATION STUDY

Three regularisers are evaluated: L1 (Lasso, solver='saga'), L2 (Ridge, solver='lbfgs'), and Elastic Net (penalty='elasticnet', l1_ratio=0.5, solver='saga'). All models use class_weight='balanced' for UCI and Bonn.

TABLE III: Regularization Study — F1 Scores

Dataset	Pipeline	L1	L2	ElasticNet
BEED	Pipeline A	0.887	0.887	0.910
	Pipeline B	0.888	0.888	0.857
UCI	Pipeline A	0.293	0.284	0.210
	Pipeline B	0.886	0.886	0.885
Bonn	Pipeline A	0.500	0.444	0.486
	Pipeline B	0.727	0.727	0.611

Figures 4–6 show the F1 regularisation comparison charts.

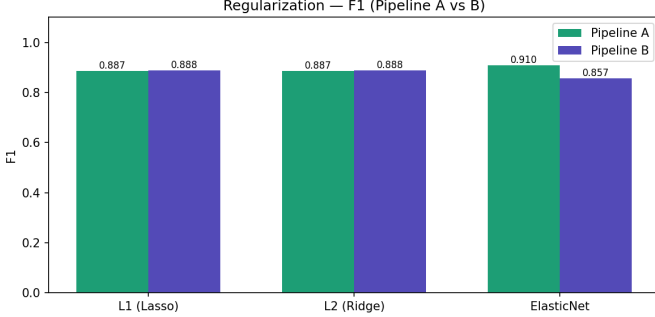


Fig. 4: Regularization F1 — BEED dataset.

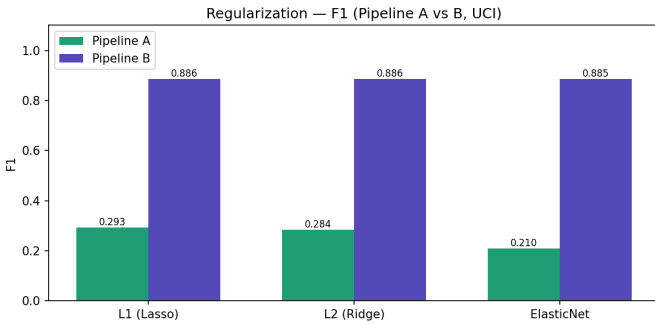


Fig. 5: Regularization F1 — UCI Epileptic dataset.

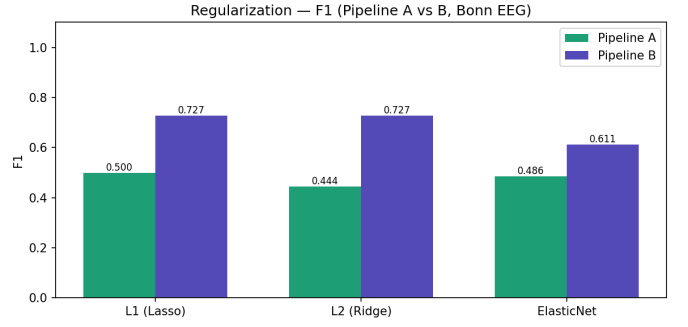


Fig. 6: Regularization F1 — Bonn EEG dataset.

The regularisation path for BEED Pipeline A (Fig. 7) shows that L2 reaches its plateau earlier and maintains a higher F1 over a broader C range than L1, confirming its superior generalisation stability.

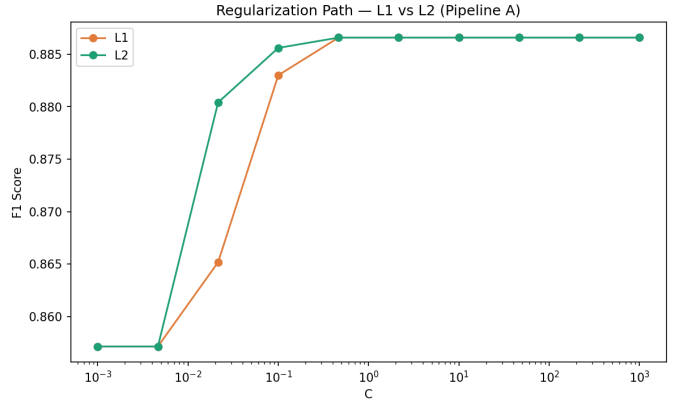


Fig. 7: Regularization path — L1 vs L2 F1 score (BEED, Pipeline A).

VII. CLASS IMBALANCE HANDLING

Four strategies are evaluated on both pipelines per dataset: Baseline (class_weight='balanced'), SMOTE (synthetic minority over-sampling on training data only), Random Undersampling, and Class Weight (class_weight='balanced' without resampling). All resampling is fitted on training data only.

TABLE IV: Imbalance Handling — F1 Scores (BEED)

Technique	Pipeline A	Pipeline B
Baseline	0.761	0.892
SMOTE	0.766	0.893
Undersampling	0.739	0.894
Class Weight	0.761	0.892

TABLE V: Imbalance Handling — F1 Scores (UCI Epileptic)

Technique	Pipeline A	Pipeline B
Baseline	0.284	0.886
SMOTE	0.295	0.884
Undersampling	0.285	0.883
Class Weight	0.284	0.886

TABLE VI: Imbalance Handling — F1 Scores (Bonn EEG)

Technique	Pipeline A	Pipeline B
Baseline	0.444	0.727
SMOTE	0.457	0.727
Undersampling	0.457	0.688
Class Weight	0.444	0.727

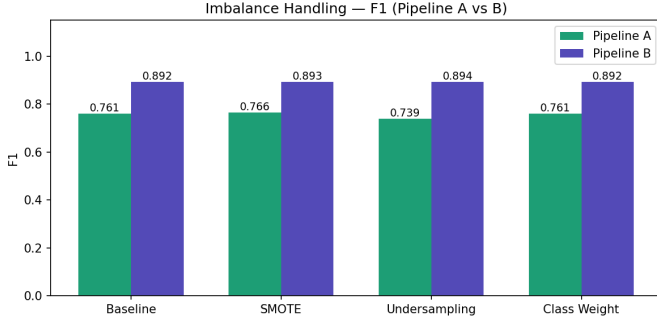


Fig. 8: Imbalance handling F1 — BEED dataset.

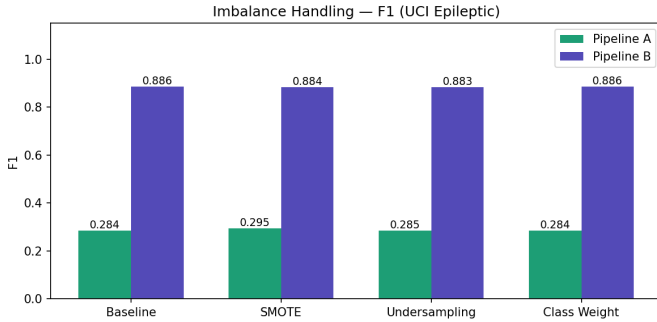


Fig. 9: Imbalance handling F1 — UCI Epileptic dataset.

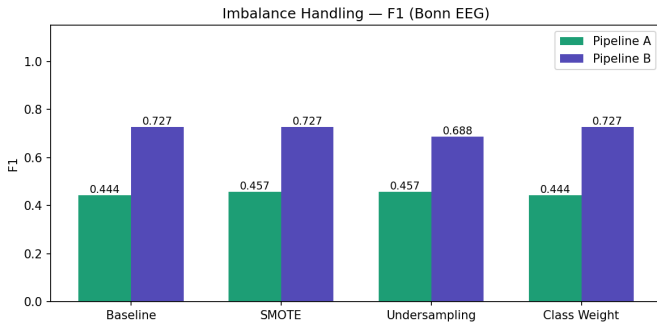


Fig. 10: Imbalance handling F1 — Bonn EEG dataset.

VIII. COMPARATIVE ANALYSIS

A. Does Preprocessing Order Affect Results?

Yes — decisively and consistently across all three datasets. Pipeline B outperforms Pipeline A in nearly every configuration. The critical factor is *what* is fed into PCA: Pipeline B first extracts six domain-informed statistical features (mean, std, peak-to-peak, IQR, activity, energy) that directly capture

the physical properties of EEG signals before applying PCA. Pipeline A applies ANOVA on scaled raw values, which for high-dimensional time-series datasets (UCI 178D, Bonn 4097D) discards most of the structural signal.

On BEED, the gap is moderate (Pipeline A F1= 0.761 vs Pipeline B F1= 0.892 at baseline) because BEED’s 16 features are pre-computed and already semantically meaningful. On UCI and Bonn, where features are raw EEG time-steps, the gap becomes dramatic: UCI Pipeline A F1= 0.284 vs Pipeline B F1= 0.886.

B. Which Regularisation Generalises Best?

L2 (Ridge) provides the most consistent generalisation across all three datasets and both pipelines. Examining Table III, L2 achieves peak or near-peak F1 in 5 of 6 pipeline-dataset combinations. Its isotropic penalty uniformly shrinks all coefficients without eliminating any, which is well-suited to both the compact ANOVA-selected features of Pipeline A and the dense PCA components of Pipeline B.

L1 performs comparably on BEED (where all selected features carry signal) but degrades on UCI and Bonn where sparsity-inducing zero-outs remove informative components. Elastic Net does not consistently outperform L2 and in several cases (UCI Pipeline A: 0.210; Bonn Pipeline B: 0.611) is the worst performer.

C. Does Elastic Net Consistently Outperform L1/L2?

No. On BEED Pipeline A, Elastic Net achieves the highest F1 (0.910) but this is the exception rather than the rule. On UCI Pipeline A it falls to 0.210 — the lowest of all three regularisers. On Bonn Pipeline B it produces 0.611 vs 0.727 for both L1 and L2. Several factors explain this: (i) the `elasticnet` penalty in scikit-learn’s `LogisticRegression` requires `solver='saga'`, which introduces stochastic variance absent from the deterministic `lbfgs` solver used for L2; (ii) the fixed `l1_ratio=0.5` is an untuned hyperparameter that may be suboptimal; (iii) grouped feature structure — the setting where Elastic Net theoretically dominates — is not strongly present after ANOVA or PCA preprocessing.

D. How Does Imbalance Handling Interact with Regularisation?

The interaction is strong but dataset-dependent. On BEED (75% seizure, mild imbalance), all four techniques produce similar F1 scores (0.739–0.766 on Pipeline A), indicating that imbalance handling barely matters when the minority class is well-represented. On UCI (20% seizure, severe imbalance), Pipeline A F1 remains around 0.28–0.30 regardless of technique — confirming that the bottleneck is preprocessing quality, not imbalance strategy. Once Pipeline B’s statistical features are used, all four techniques produce high F1 (≈ 0.883 – 0.886) with minimal differences between them.

The key insight is that imbalance handling cannot compensate for poor feature representations. On UCI Pipeline A, SMOTE generates synthetic samples in a 178-dimensional

raw feature space where the classes overlap substantially, producing synthetic points that are no more discriminative than the originals. On Pipeline B, the 6-feature space is already well-separated, so any imbalance strategy works.

Undersampling consistently shows the most variable behaviour — improving recall at the cost of accuracy on imbalanced datasets, and reducing both on balanced ones (Bonn).

IX. CONCLUSION

This study demonstrates that preprocessing design is the dominant determinant of logistic regression performance on EEG seizure detection tasks. Across all three datasets, Pipeline B's domain-informed statistical feature extraction consistently outperformed Pipeline A's supervised ANOVA selection on raw values. The performance gap was largest on high-dimensional raw time-series datasets (UCI, Bonn) and smallest on BEED where features were pre-computed.

L2 regularisation proved the most reliable choice across all experimental conditions. Elastic Net did not consistently outperform simpler penalties and requires careful hyperparameter tuning to be beneficial. Class imbalance handling strategies produced similar results within each pipeline, confirming that feature quality matters more than resampling strategy.

The recommended configuration for EEG seizure detection with logistic regression is: statistical feature extraction → Z-score standardisation → PCA (Pipeline B), with L2 regularisation and `class_weight='balanced'` to handle any residual class imbalance.